

Large Language Models for Generative Recommendation and User Behavior Modeling: A Survey

Large language models are reshaping recommender systems by turning recommendation from score prediction over fixed identifiers into language-conditioned generation, reasoning over item content, conversational preference elicitation, and user behavior modeling. At the same time, recommender systems supply demanding tests for LLMs: they require personalization, temporal adaptation, calibrated ranking, privacy-aware user modeling, and evaluation under feedback loops. This survey synthesizes an arXiv-derived corpus of 300 papers collected during this run and organizes the literature around LLM-based recommender models, generative recommendation, sequential and session behavior modeling, user preference modeling, conversational recommendation, multimodal and knowledge-enhanced recommendation, ranking pipelines, and evaluation. The survey argues that the next generation of recommendation models must connect natural-language semantics with user-item interaction structure, long-term preference dynamics, and deployment-aware safeguards.

Keywords:

large language models; recommender systems; generative recommendation; user behavior modeling; personalization; sequential recommendation; conversational recommendation; ranking

1 Introduction

Recommendation has traditionally been framed as predicting a user’s affinity for candidate items from interaction history, item features, and contextual signals. Large language models (LLMs) broaden this framing. They can parse user requests, reason over item descriptions and reviews, generate explanations, ask clarifying questions, summarize preferences, and represent behavior histories as natural-language or token sequences. This shift makes recommendation a bridge between information retrieval, sequence modeling, personalization, natural-language generation, and human-AI interaction [41, 75, 97, 140, 165, 169, 174, 299].

The phrase *generative recommendation* covers several related ideas. A model may generate item identifiers, queries, item descriptions, ranked lists, conversational responses, or preference explanations. It may also treat a user history as a sequence and autoregressively predict the next item, or it may use an LLM to transform heterogeneous content into

representations for retrieval and ranking. User behavior modeling is the companion problem: the system must infer short-term intent, long-term interests, preference drift, context, and uncertainty from clicks, purchases, ratings, reviews, sessions, and language feedback [38, 43, 64, 76, 113, 152, 294, 298].

This survey reviews LLMs for generative recommendation and user behavior modeling. We focus on the methodological interface between language models and recommender systems: how item and user signals are represented, how generation is constrained into valid recommendations, how behavior sequences are modeled, how conversational feedback changes user profiles, and how evaluation should account for ranking quality, personalization, trust, fairness, and privacy.

2 Background

2.1 Recommender Systems as User-Item Interaction Models

Recommender systems learn from user-item interactions such as clicks, ratings, purchases, dwell time, reviews, and session transitions. Classical collaborative filtering uses interaction matrices; neural recommenders add representation learning, graph structure, attention, and deep ranking models. Modern pipelines usually separate candidate retrieval, ranking, reranking, explanation, and feedback collection. LLM-based methods can enter any of these stages, but their benefits and risks differ depending on whether they operate over item identifiers, natural-language content, behavior sequences, or dialogue state [38, 41, 97, 140, 162, 226, 255, 299].

2.2 Why Language Models Matter for Recommendation

Recommendation data is increasingly textual and multimodal. Items have titles, descriptions, reviews, metadata, images, transcripts, and community discussions; users express intent through queries, conversations, critiques, and free-form feedback. LLMs provide a shared semantic interface for these signals. They can reduce cold-start friction by reading item content, support zero-shot or few-shot recommendation, generate explanations, and mediate conversations. However, LLMs do not automatically solve collaborative filtering: popularity bias, temporal dynamics, sparse feedback, and ranking calibration remain core recommender-system problems [41, 75, 97, 140, 165, 169, 174, 299].

2.3 From Sequential Recommendation to Generative Modeling

Sequential recommenders treat a user’s history as a behavior sequence and predict the next item or next basket. Transformer architectures made this setting close to language modeling: items become tokens, histories become sequences, and next-item prediction resembles next-token prediction. LLM-era work extends this analogy by mixing item identifiers with natural language, prompts, reviews, and user profiles. The challenge is to preserve the precision of ranking while exploiting the semantic flexibility of language generation [55, 86, 165, 182, 226, 252, 255, 295].

3 Methods

3.1 Search Protocol and Corpus Construction

The corpus for this survey was generated during this run from the arXiv structured API. Queries combined the task topic with large language models, LLM recommenders, generative recommendation, generative retrieval, user behavior modeling, user preference modeling, personalized recommendation, sequential and session-based recommendation, BERT4Rec, SASRec, P5-style recommendation, TALLRec, RecGPT, LLMRec, GPT4Rec, conversational recommendation, ChatGPT, prompt learning, instruction tuning, ranking, reranking, CTR prediction, collaborative filtering, item descriptions, review-aware recommendation, knowledge-graph recommendation, multimodal recommendation, news recommendation, POI recommendation, cold-start recommendation, explainability, and recommendation benchmarks. Records were retained when title or abstract evidence connected recommendation or user behavior modeling with LLMs, language models, generative modeling, transformers, conversational systems, or behavior-sequence modeling. Duplicates were removed by arXiv identifier, and the final bibliography cites 300 unique scholarly preprints.

3.2 Taxonomy

We organize the literature along eight axes. The first is *model role*: retriever, ranker, generator, explainer, conversational agent, user simulator, or profile encoder. The second is *input signal*: item identifiers, interactions, sessions, ratings, reviews, queries, item text, images, knowledge graphs, or dialogue turns. The third is *output object*: item ID, ranked list, natural-language response, explanation, critique, synthetic behavior, or latent user representation. The fourth is *training signal*: next-item prediction, contrastive learning, masked modeling, supervised instruction tuning, preference optimization, reinforcement learning, or distillation. The fifth is *personalization horizon*: immediate intent, session context, long-term interests, cross-domain transfer, or evolving preference. The sixth is *pipeline stage*: candidate generation, matching, ranking, reranking, explanation, and feedback. The seventh is *evaluation*: accuracy, ranking quality, calibration, diversity, novelty, explanation quality, conversational success, latency, and cost. The eighth is *risk*: hallucinated items, privacy leakage, manipulation, popularity bias, unfair exposure, and feedback-loop instability.

3.3 LLM-Based Recommender Architectures

LLM recommenders use language models in multiple roles. Some prompt a frozen LLM with user histories and item descriptions. Others tune the model on recommendation instructions, map item IDs into language-model token spaces, or distill LLM semantics into smaller retrievers and rankers. A practical architecture often combines an efficient candidate generator with an LLM-based reranker or explainer, because full-list generation by a large model can be costly and difficult to calibrate [41, 75, 97, 140, 165, 169, 174, 299].

3.4 Generative Recommendation

Generative methods formulate recommendation as producing the next item, a set of candidate items, a text query, or a ranked response. Identifier generation can exploit autoregressive sequence modeling but must avoid invalid item tokens and popularity collapse. Text generation can explain preferences or create queries for retrieval, but it must remain grounded in an item catalog. Generative retrieval connects these views by learning to map user intent into item or document identifiers [38, 43, 64, 76, 113, 152, 294, 298].

3.5 Sequential and Session Behavior Modeling

Behavior-sequence models capture temporal patterns in clicks, purchases, listens, views, and searches. Transformer recommendation models learn attention over past interactions, while LLM-based variants add textual semantics and richer context. Important design choices include how to tokenize items, how to represent time gaps, how to combine short-term session intent with long-term preferences, and how to update profiles after new feedback [55, 86, 165, 182, 226, 252, 255, 295].

3.6 User Preference Modeling and Personalization

Personalization requires more than memorizing historical items. Systems must infer what attributes matter to the user, which contexts change intent, when preferences drift, and when uncertainty should trigger exploration or clarification. LLMs can summarize user histories into profiles and reason about constraints, but summaries can overfit, expose sensitive traits, or erase minority interests. Strong systems keep provenance, uncertainty, and editable user

control in the loop [41, 75, 140, 165, 169, 174, 255, 299].

3.7 Conversational and Interactive Recommendation

Conversational recommendation turns ranking into an interactive process. The system can ask questions, accept critiques, explain trade-offs, and revise candidate sets. LLMs make such dialogue fluent, but fluency is not equivalent to useful recommendation. The model must connect conversation state to catalog retrieval, respect constraints, avoid hallucinated items, and learn from explicit feedback without manipulating the user [41, 48, 75, 97, 169, 174, 177, 299].

3.8 Multimodal, Review-Aware, and Knowledge-Enhanced Recommendation

Many domains require content beyond IDs: movies have plots and trailers, products have images and reviews, papers have abstracts and citation graphs, and venues have locations and social context. LLMs and multimodal models can encode this side information, while knowledge graphs provide structured relations among users, items, attributes, and entities. These signals are especially valuable for cold-start, explanation, cross-domain transfer, and long-tail items [61, 108, 140, 165, 169, 182, 226, 244].

3.9 Ranking, CTR Prediction, and Pipeline Integration

Industrial recommendation pipelines remain ranking systems under latency, freshness, and business constraints. LLMs can support candidate generation, feature construction, reranking, explanation, and synthetic data generation, but final deployment often needs compact models, caching, retrieval indexes, and calibrated ranking objectives. CTR prediction and reranking papers remind us that generative capability must be reconciled with exposure control, click feedback, and counterfactual evaluation [38, 41, 97, 140, 162, 226, 255, 299].

3.10 Evaluation and Benchmarks

Evaluation is unusually subtle. Offline metrics such as Recall at K and NDCG do not fully measure conversational quality, explanation faithfulness, user satisfaction, or long-term feedback loops. LLM evaluations add risks of invalid generated items, leakage from pretraining data, prompt sensitivity, and expensive inference. Recommendation benchmarks should

report catalog validity, temporal splits, cold-start settings, diversity, fairness, privacy, latency, and user-facing utility [41, 75, 97, 140, 165, 169, 174, 299].

3.11 Representative Literature Map

The following map cites the retained corpus by theme, making the coverage explicit across LLM recommender models, generative recommendation, sequential behavior modeling, personalization, conversational recommendation, multimodal and knowledge-enhanced recommendation, ranking pipelines, and evaluation.

Llm-based and foundation recommender models

The llm-based and foundation recommender models cluster contains work that develops methods, systems, datasets, or analyses central to this survey [55, 93, 102, 113, 128, 162, 226, 289, 298]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [2, 17, 88, 121, 127, 139, 203, 219, 267]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [29, 66, 79, 109, 157, 199, 237, 247, 291]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [26, 69, 106, 160, 166, 193, 213, 268, 282]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [25, 73, 85, 90, 125, 126, 156, 251, 259]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [9, 94, 112, 115, 141, 149, 216, 265, 300]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [269].

Generative recommendation and item generation

The generative recommendation and item generation cluster contains work that develops methods, systems, datasets, or analyses central to this survey [14, 76, 80, 116, 118, 123, 171, 197, 263]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [172].

Sequential, session-based, and behavior-sequence modeling The sequential, session-based, and behavior-sequence modeling cluster contains work that develops methods, systems,

datasets, or analyses central to this survey [84, 86, 103, 122, 124, 207, 254, 285, 295]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [23, 82, 104, 114, 117, 209, 239, 248, 278]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [70, 132, 134, 150, 185, 195, 210, 238, 292]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [13, 15, 27, 60, 77, 95, 208, 297].

User preference modeling and personalization

The user preference modeling and personalization cluster contains work that develops methods, systems, datasets, or analyses central to this survey [19, 32, 50, 64, 71, 72, 165, 260, 293]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [10, 44, 96, 196, 229, 245, 249, 262, 272]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [22, 45, 58, 137, 178, 179, 198, 206, 228]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [176, 258].

Conversational and interactive recommendation

The conversational and interactive recommendation cluster contains work that develops methods, systems, datasets, or analyses central to this survey [43, 46, 48, 52, 75, 97, 169, 174, 299]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [12, 18, 131, 142, 194, 212, 220, 221, 286]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [37, 59, 107, 167, 191, 192, 231, 243, 256]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [63, 120, 138, 211, 230, 233, 264, 271, 283]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [53, 62, 155].

Multimodal, review-aware, and knowledge-enhanced recommendation The multimodal, review-aware, and knowledge-enhanced recommendation cluster contains work that develops methods, systems, datasets, or analyses central to this survey [21, 54,

56, 61, 89, 168, 182, 188, 257]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [8, 83, 91, 136, 225, 235, 274, 281, 288]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [148, 159, 200, 201, 205, 218, 223, 280, 284]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [24, 40, 65, 92, 130, 145, 187, 214, 261]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [3, 51, 100, 161, 204, 234, 240, 246, 250].

Ranking, ctr prediction, and recommender-system pipelines The ranking, ctr prediction, and recommender-system pipelines cluster contains work that develops methods, systems, datasets, or analyses central to this survey [7, 38, 41, 105, 140, 244, 252, 255, 294]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [5, 16, 34, 42, 57, 98, 101, 236, 287]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [35, 49, 67, 144, 173, 222, 242, 275, 277]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [78, 87, 119, 184, 276].

Benchmarks, evaluation, fairness, privacy, and robustness The benchmarks, evaluation, fairness, privacy, and robustness cluster contains work that develops methods, systems, datasets, or analyses central to this survey [108, 147, 151, 152, 177, 186, 215, 224, 279]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [28, 33, 68, 110, 143, 163, 217, 227, 266]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [1, 20, 111, 135, 154, 170, 190, 202, 290]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [4, 6, 36, 47, 74, 158, 181, 241, 273]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions

[39, 81, 99, 129, 146, 164, 183, 232, 270]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [133].

4 Challenges and Prospects

4.1 Catalog-Grounded Generation

LLMs can produce persuasive recommendations that are not in the catalog or do not satisfy constraints. Future systems need constrained decoding, retrieval-grounded generation, item-validity checks, and catalog-aware feedback loops so that language generation remains tied to real recommendable objects.

4.2 Preference Dynamics and Memory

User interests change over minutes, months, and life stages. LLM recommenders need memory mechanisms that distinguish stable preferences from transient context, record provenance for inferred traits, and allow users to inspect or edit profiles.

4.3 Efficient Hybrid Pipelines

Full LLM inference over large candidate sets is often impractical. A likely direction is hybrid design: efficient retrievers and rankers produce candidates, while LLMs interpret intent, enrich representations, rerank small sets, generate explanations, or collect feedback.

4.4 Evaluation Under Feedback Loops

Recommendation affects what users see, which affects future data. LLM-based recommenders add conversational feedback and synthetic reasoning traces to this loop. Evaluation must therefore move beyond static test sets toward temporal, counterfactual, interactive, and user-centered protocols.

4.5 Trust, Privacy, and Fair Exposure

User modeling can infer sensitive traits. Natural-language explanations can persuade without being faithful. Exposure can amplify popularity and demographic bias. Future LLM recommenders should include privacy-preserving profiles, audit logs, fairness-aware ranking, and controls that make personalization legible to users.

5 Conclusion

Large language models give recommender systems a new semantic and generative interface, but they do not remove the core problems of user behavior modeling, ranking, calibration, and deployment. This survey reviewed the literature across LLM-based recommendation, generative retrieval, sequential and session modeling, personalization, conversational interaction, multimodal and knowledge-enhanced recommendation, ranking pipelines, and evaluation. The central lesson is that strong LLM recommenders will be hybrid systems: grounded in catalogs and interaction data, fluent in language, aware of preference dynamics, and evaluated against both ranking quality and human trust.

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